

Class 8

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Background

In today's class we will apply the methods and techniques clustering and PCA to help make sense of a real world breast cancer FNA biopsy data set.

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names = 1)
```

```
head(wisc.df)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
84358402	M	20.29	14.34	135.10	1297.0
843786	M	12.45	15.70	82.57	477.1

	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean
842302	0.11840	0.27760	0.3001	0.14710
842517	0.08474	0.07864	0.0869	0.07017
84300903	0.10960	0.15990	0.1974	0.12790
84348301	0.14250	0.28390	0.2414	0.10520
84358402	0.10030	0.13280	0.1980	0.10430
843786	0.12780	0.17000	0.1578	0.08089

	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
84358402	0.1809	0.05883	0.7572	0.7813	5.438
843786	0.2087	0.07613	0.3345	0.8902	2.217

	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
84358402	94.44	0.011490	0.02461	0.05688	0.01885
843786	27.19	0.007510	0.03345	0.03672	0.01137

	symmetry_se	fractal_dimension_se	radius_worst	texture_worst
842302	0.03003	0.006193	25.38	17.33
842517	0.01389	0.003532	24.99	23.41
84300903	0.02250	0.004571	23.57	25.53
84348301	0.05963	0.009208	14.91	26.50
84358402	0.01756	0.005115	22.54	16.67
843786	0.02165	0.005082	15.47	23.75

	perimeter_worst	area_worst	smoothness_worst	compactness_worst
842302	184.60	2019.0	0.1622	0.6656
842517	158.80	1956.0	0.1238	0.1866
84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
84358402	152.20	1575.0	0.1374	0.2050
843786	103.40	741.6	0.1791	0.5249

	concavity_worst	concave.points_worst	symmetry_worst
842302	0.7119	0.2654	0.4601
842517	0.2416	0.1860	0.2750
84300903	0.4504	0.2430	0.3613
84348301	0.6869	0.2575	0.6638
84358402	0.4000	0.1625	0.2364
843786	0.5355	0.1741	0.3985

	fractal_dimension_worst
842302	0.11890
842517	0.08902
84300903	0.08758
84348301	0.17300
84358402	0.07678
843786	0.12440

```
wisc.data <- wisc.df[,-1]
diagnosis <- wisc.df$diagnosis
head(wisc.data)
```

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean
842302	17.99	10.38	122.80	1001.0	0.11840

842517	20.57	17.77	132.90	1326.0	0.08474
84300903	19.69	21.25	130.00	1203.0	0.10960
84348301	11.42	20.38	77.58	386.1	0.14250
84358402	20.29	14.34	135.10	1297.0	0.10030
843786	12.45	15.70	82.57	477.1	0.12780
compactness_mean concavity_mean concave.points_mean symmetry_mean					
842302	0.27760	0.3001		0.14710	0.2419
842517	0.07864	0.0869		0.07017	0.1812
84300903	0.15990	0.1974		0.12790	0.2069
84348301	0.28390	0.2414		0.10520	0.2597
84358402	0.13280	0.1980		0.10430	0.1809
843786	0.17000	0.1578		0.08089	0.2087
fractal_dimension_mean radius_se texture_se perimeter_se area_se					
842302		0.07871	1.0950	0.9053	8.589 153.40
842517		0.05667	0.5435	0.7339	3.398 74.08
84300903		0.05999	0.7456	0.7869	4.585 94.03
84348301		0.09744	0.4956	1.1560	3.445 27.23
84358402		0.05883	0.7572	0.7813	5.438 94.44
843786		0.07613	0.3345	0.8902	2.217 27.19
smoothness_se compactness_se concavity_se concave.points_se					
842302	0.006399	0.04904	0.05373		0.01587
842517	0.005225	0.01308	0.01860		0.01340
84300903	0.006150	0.04006	0.03832		0.02058
84348301	0.009110	0.07458	0.05661		0.01867
84358402	0.011490	0.02461	0.05688		0.01885
843786	0.007510	0.03345	0.03672		0.01137
symmetry_se fractal_dimension_se radius_worst texture_worst					
842302	0.03003	0.006193	25.38		17.33
842517	0.01389	0.003532	24.99		23.41
84300903	0.02250	0.004571	23.57		25.53
84348301	0.05963	0.009208	14.91		26.50
84358402	0.01756	0.005115	22.54		16.67
843786	0.02165	0.005082	15.47		23.75
perimeter_worst area_worst smoothness_worst compactness_worst					
842302	184.60	2019.0	0.1622		0.6656
842517	158.80	1956.0	0.1238		0.1866
84300903	152.50	1709.0	0.1444		0.4245
84348301	98.87	567.7	0.2098		0.8663
84358402	152.20	1575.0	0.1374		0.2050
843786	103.40	741.6	0.1791		0.5249
concavity_worst concave.points_worst symmetry_worst					
842302	0.7119		0.2654		0.4601
842517	0.2416		0.1860		0.2750

84300903	0.4504	0.2430	0.3613
84348301	0.6869	0.2575	0.6638
84358402	0.4000	0.1625	0.2364
843786	0.5355	0.1741	0.3985
fractal_dimension_worst			
842302	0.11890		
842517	0.08902		
84300903	0.08758		
84348301	0.17300		
84358402	0.07678		
843786	0.12440		

Make sure to remove the first diagnosis column. I don't want to use this for my machine learning models. We will use it later to compare our results to expert diagnosis

```
wisc.data <- wisc.df[,-1]
diagnosis <- wisc.df$diagnosis
head(wisc.data)
```

	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	
842302	17.99	10.38	122.80	1001.0	0.11840	
842517	20.57	17.77	132.90	1326.0	0.08474	
84300903	19.69	21.25	130.00	1203.0	0.10960	
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84358402	20.29	14.34	135.10	1297.0	0.10030	
843786	12.45	15.70	82.57	477.1	0.12780	
	compactness_mean	concavity_mean	concave.points_mean	symmetry_mean		
842302	0.27760	0.3001		0.14710	0.2419	
842517	0.07864	0.0869		0.07017	0.1812	
84300903	0.15990	0.1974		0.12790	0.2069	
84348301	0.28390	0.2414		0.10520	0.2597	
84358402	0.13280	0.1980		0.10430	0.1809	
843786	0.17000	0.1578		0.08089	0.2087	
	fractal_dimension_mean	radius_se	texture_se	perimeter_se	area_se	
842302		0.07871	1.0950	0.9053	8.589	153.40
842517		0.05667	0.5435	0.7339	3.398	74.08
84300903		0.05999	0.7456	0.7869	4.585	94.03
84348301		0.09744	0.4956	1.1560	3.445	27.23
84358402		0.05883	0.7572	0.7813	5.438	94.44
843786		0.07613	0.3345	0.8902	2.217	27.19
	smoothness_se	compactness_se	concavity_se	concave.points_se		
842302	0.006399	0.04904	0.05373	0.01587		

842517	0.005225	0.01308	0.01860	0.01340
84300903	0.006150	0.04006	0.03832	0.02058
84348301	0.009110	0.07458	0.05661	0.01867
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84300903	152.50	1709.0	0.1444	0.4245
84348301	98.87	567.7	0.2098	0.8663
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	concavity_worst	concave.points_worst	symmetry_worst	
842302	0.7119	0.2654	0.4601	
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84300903	0.4504	0.2430	0.3613	
84348301	0.6869	0.2575	0.6638	
84358402	0.4000	0.1625	0.2364	
843786	0.5355	0.1741	0.3985	
	fractal_dimension_worst			
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842517	0.08902			
84300903	0.08758			
84348301	0.17300			
84358402	0.07678			
843786	0.12440			

Q1. How many observations are in this dataset?

```
nrow(wisc.df)
```

```
[1] 569
```

The dataset contains 569 rows of data.

Q2. How many of the observations have a malignant diagnosis?

```
sum(diagnosis=="M")
```

```
[1] 212
```

There are 212 observations with a malignant diagnosis.

```
table(wisc.df$diagnosis)
```

```
  B   M  
357 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

```
length(grep("_mean$", colnames(wisc.df)))
```

```
[1] 10
```

10 are suffixed with `_mean`

##Principal Component Analysis

The main function here is `prcomp()` and we want to make sure we set the optional argument `scale=TRUE`:

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)  
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010

Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

PC1 accounts for roughly 44% of the original variance in the dataset.

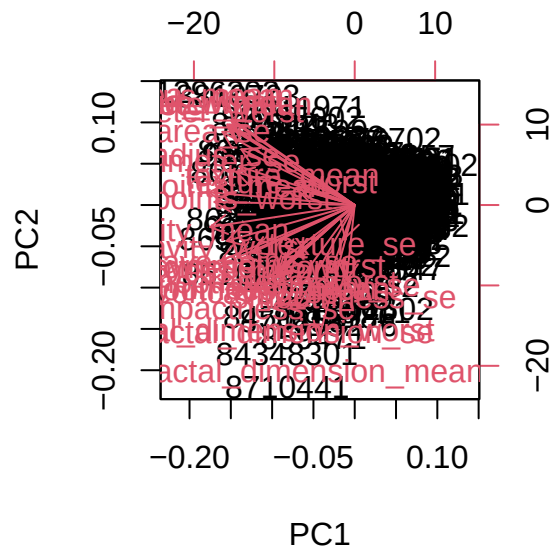
Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

Capturing 70% or more of the original variability requires three principal components.

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

The first seven principal components are needed to account for at least 90% of the original variance in the dataset.

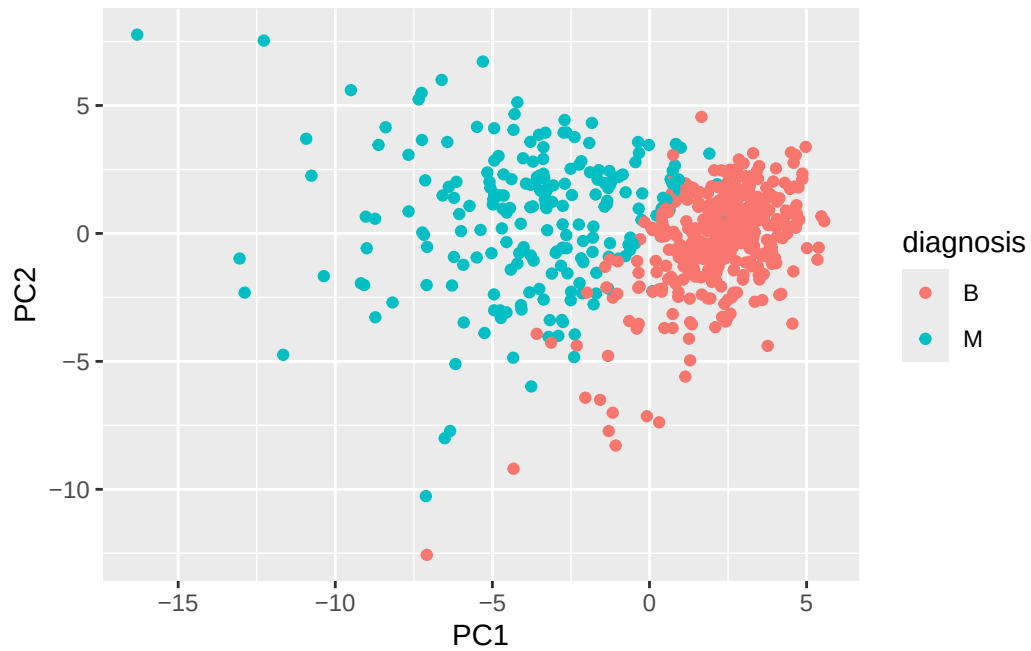
```
library(ggplot2)
biplot(wisc.pr)
```



Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

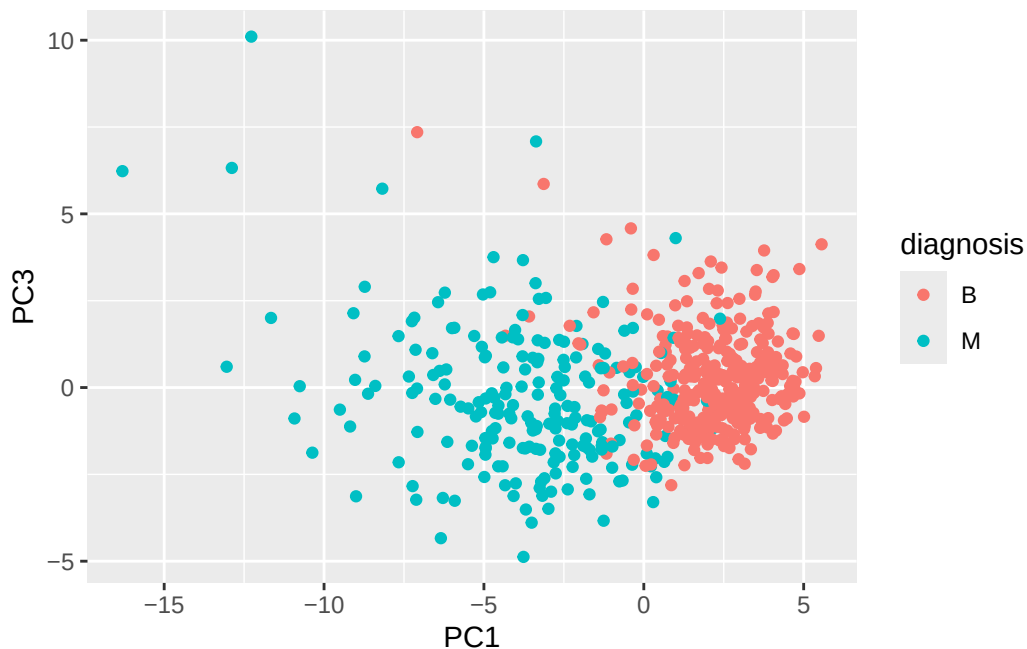
What immediately stands out is how crowded and cluttered the plot appears. With so many variables overlapping, it becomes difficult to interpret and visually overwhelming.

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +  
aes(PC1, PC3, col=diagnosis) +  
geom_point()
```



Collectively these two plots (“score plot” and “loadings plot”) tells us that if cells nucleus are deeply indented (“concave”), irregular and non circular (“compactness”), and have larger “perimeter” values they tend to be malignant

Q.9 For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

```
sort(abs(wisc.pr$rotation[, 1]), decreasing = TRUE)
```

concave.points_mean	concavity_mean	concave.points_worst
0.26085376	0.25840048	0.25088597
compactness_mean	perimeter_worst	concavity_worst
0.23928535	0.23663968	0.22876753
radius_worst	perimeter_mean	area_worst
0.22799663	0.22753729	0.22487053
area_mean	radius_mean	perimeter_se

0.22099499	0.21890244	0.21132592
compactness_worst	radius_se	area_se
0.21009588	0.20597878	0.20286964
concave.points_se	compactness_se	concavity_se
0.18341740	0.17039345	0.15358979
smoothness_mean	symmetry_mean	fractal_dimension_worst
0.14258969	0.13816696	0.13178394
smoothness_worst	symmetry_worst	texture_worst
0.12795256	0.12290456	0.10446933
texture_mean	fractal_dimension_se	fractal_dimension_mean
0.10372458	0.10256832	0.06436335
symmetry_se	texture_se	smoothness_se
0.04249842	0.01742803	0.01453145

concave.points_mean has one of the strongest loadings on PC1, indicating it contributes heavily to the first principal component. When comparing absolute loadings, no other feature shows a clearly larger contribution.

Hierarchical clustering

First scale the data (with the scale function), then calculate a distance matrix (with the dist() function) Then cluster with the hclust() function and plot:

```
wisc.hclut <- hclust(dist( scale(wisc.data)))
```

Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclut)
```

Cluster Dendrogram



```
dist(scale(wisc.data))  
hclust(*, "complete")
```

You can also use:

```
wisc.hclut.clusters <- cutree(wisc.hclut, k = 4)  
table(wisc.hclut.clusters)
```

```
wisc.hclut.clusters  
  1  2  3  4  
177  7 383  2
```

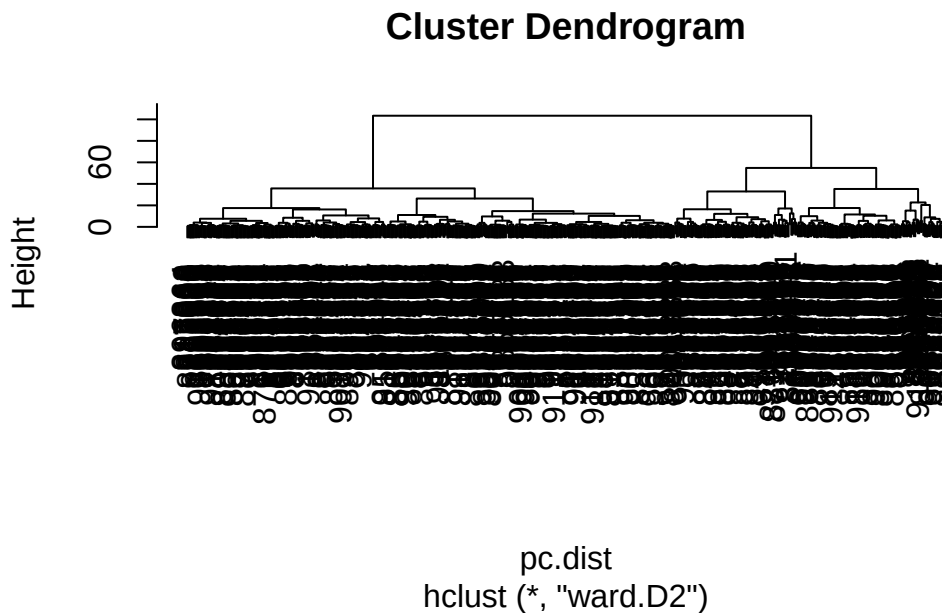
Q12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

I prefer the the PCA-based visualizations give the clearer and more interpretable results.

Combining methods

Here we will tke our PCA results and use those as input for clustering. In other words wisc.pr\$x scores that we plotted aboce (the main output from our PCA - how the data lie on our new principal componet axis/variables) and use a subset pf thes PCs that capture the most variance as input for hclut()

```
pc.dist <- dist(wisc.pr$x[,1:3])
wisc.pr.hclust <- hclust(pc.dist, method = "ward.D2")
plot(wisc.pr.hclust)
```



```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1  2
203 366
```

Q13. How well does the newly created clust model with two clusters separate out the two “M” and “B” diagnoses?

I want to determine how the cluster assignments (groups 1 and 2) align with the expert diagnostic labels.

```
table(grps,diagnosis)
```

```
      diagnosis
grps   B    M
  1  24 179
  2 333  33
```

Cluster 1 is dominated by malignant cases (179), while Cluster 2 is primarily composed of benign cases (333).

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

```
table(wisc.hclut.clusters, diagnosis)
```

	diagnosis	
wisc.hclut.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Hierarchical clustering without PCA performs noticeably worse, since it mixes malignant and benign cases much more than the PCA-based clustering.

Sensitivity

Sensitivity $TP/(TP+FN)$

```
179/(179+33)
```

```
[1] 0.8443396
```

Sensitivity $TN/(TN+FP)$

```
333/(333+24)
```

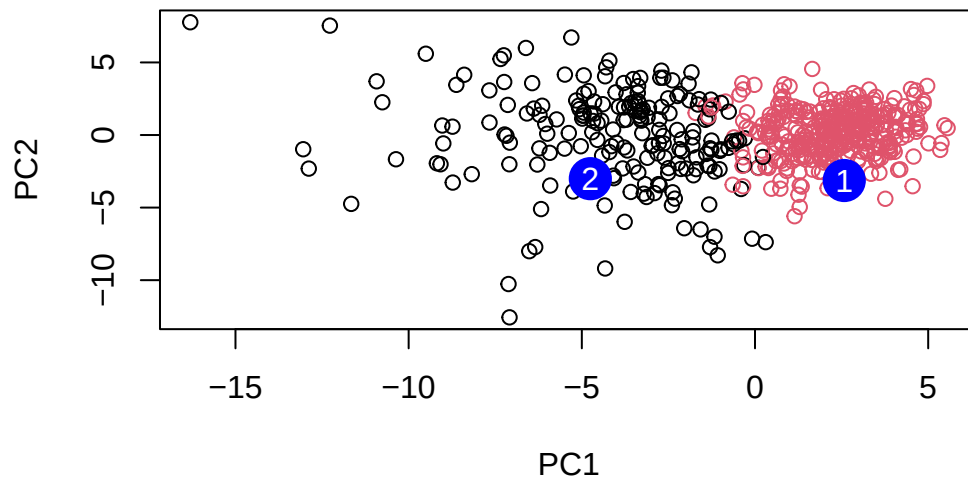
```
[1] 0.9327731
```

Prediction

```
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Patient 1 should be prioritized because their profile clusters predominantly with malignant samples, whereas Patient 2 clusters mainly with benign samples.